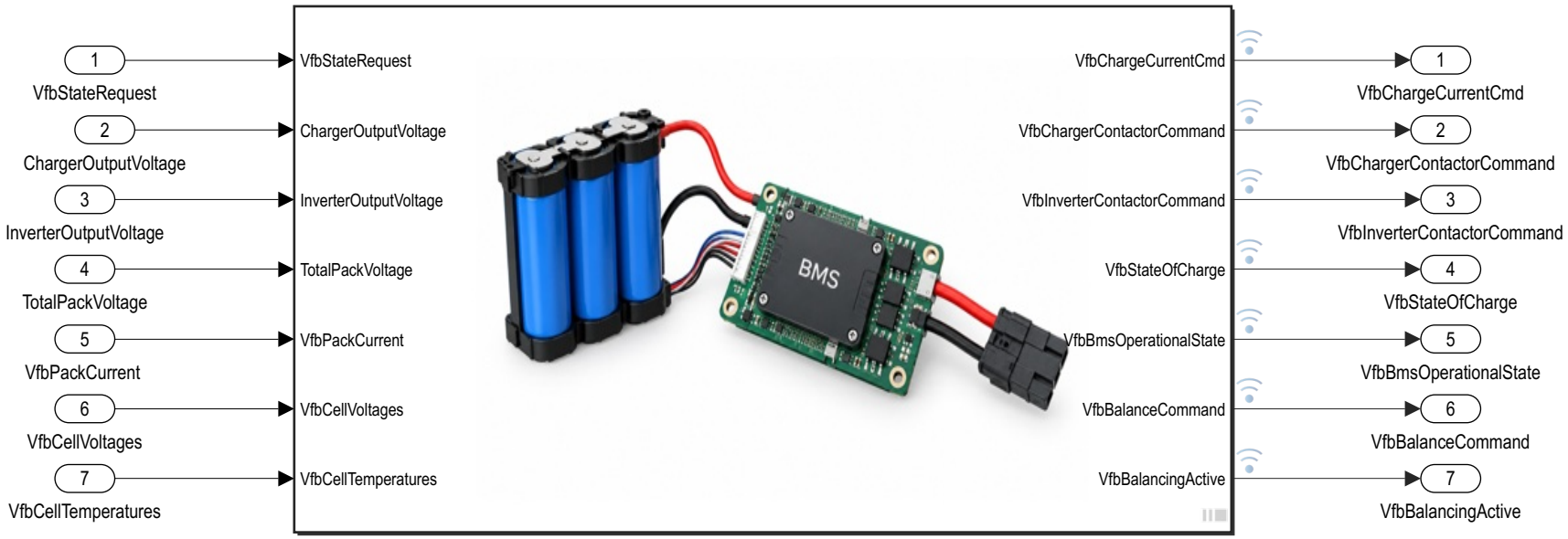
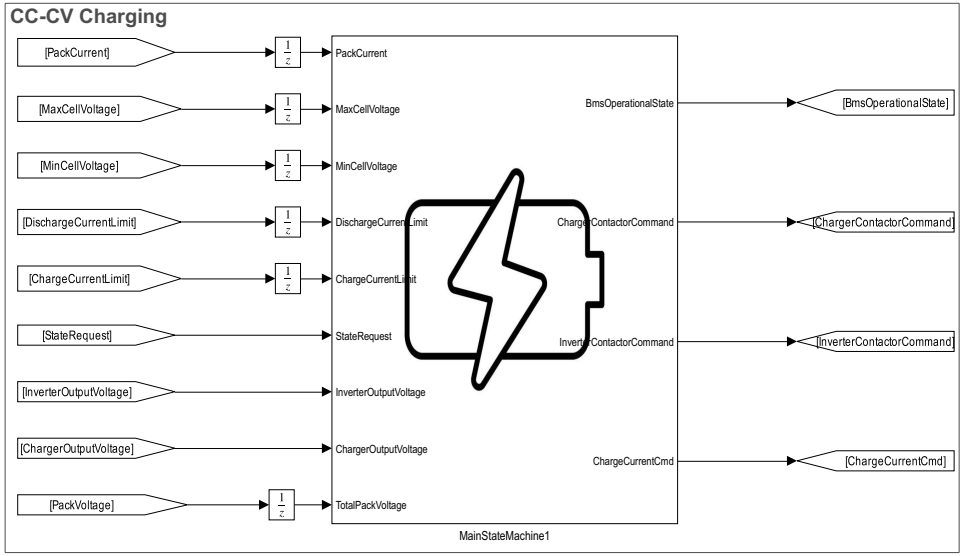
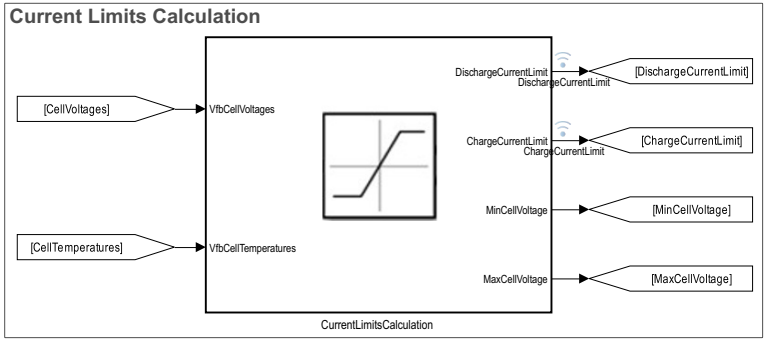
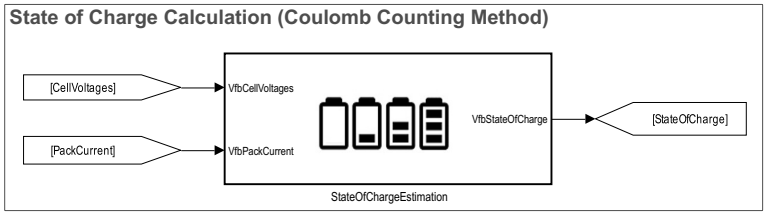
BatteryManagementSystem_V1

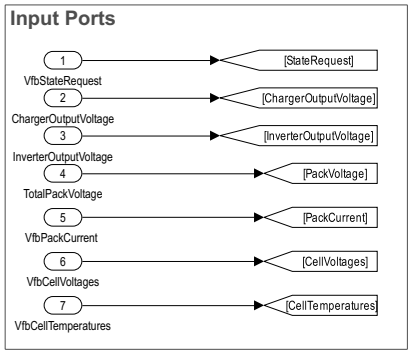


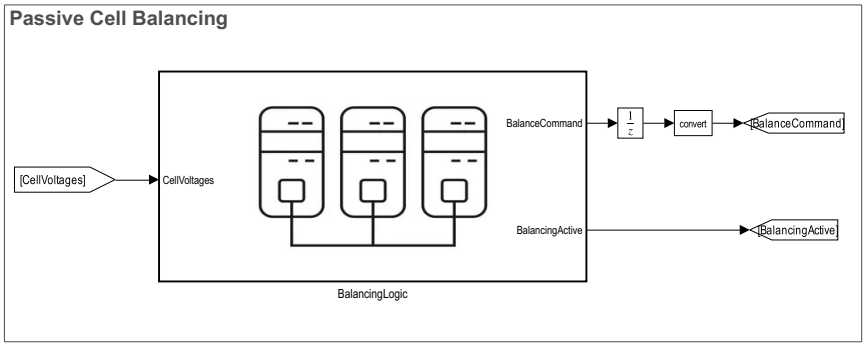
BatteryManagementSystem

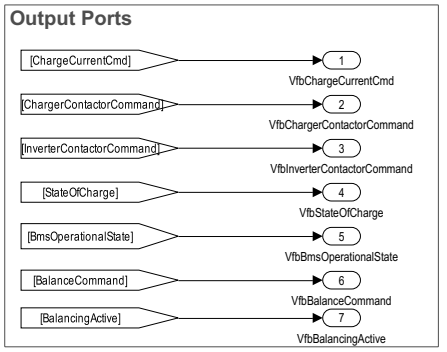

CC-CV Charging
[PackCurrent]
[MaxCellVoltage]
[MinCellVoltage]
[DischargeCurrentLimit]
[ChargeCurrentLimit]
[StateRequest]
[InverterOutputVoltage]
[ChargerOutputVoltage]
[PackVoltage]
PackCurrent
MaxCellVoltage
MinCellVoltage
DischargeCurrentLimit
ChargeCurrentLimit
StateRequest
InverterOutputVoltage
ChargerOutputVoltage
TotalPackVoltage
BmsOperationalState
ChargerContactorCommand
InverterContactorCommand
ChargeCurrentCmd
[BmsOperationalState]
[ChargerContactorCommand]
[InverterContactorCommand]
[ChargeCurrentCmd]
MainStateMachine1

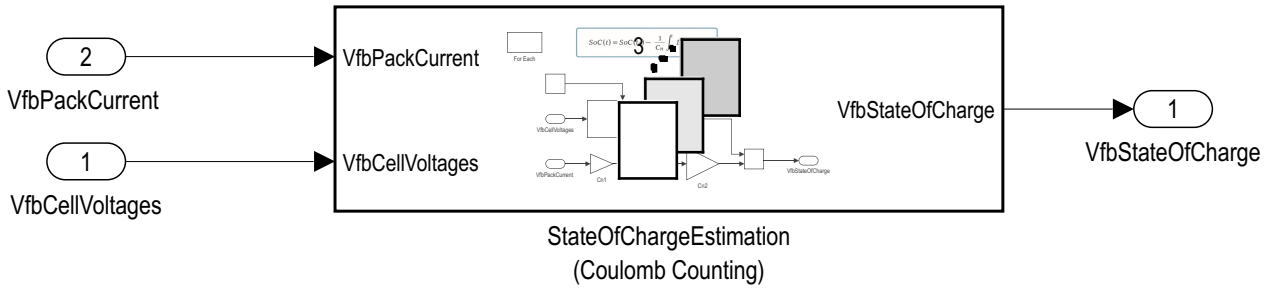

Current Limits Calculation
[CellVoltages]
[CellTemperatures]
VfbCellVoltages
VfbCellTemperatures
DischargeCurrentLimit
ChargeCurrentLimit
MinCellVoltage
MaxCellVoltage
[DischargeCurrentLimit]
[ChargeCurrentLimit]
[MinCellVoltage]
[MaxCellVoltage]
CurrentLimitsCalculation

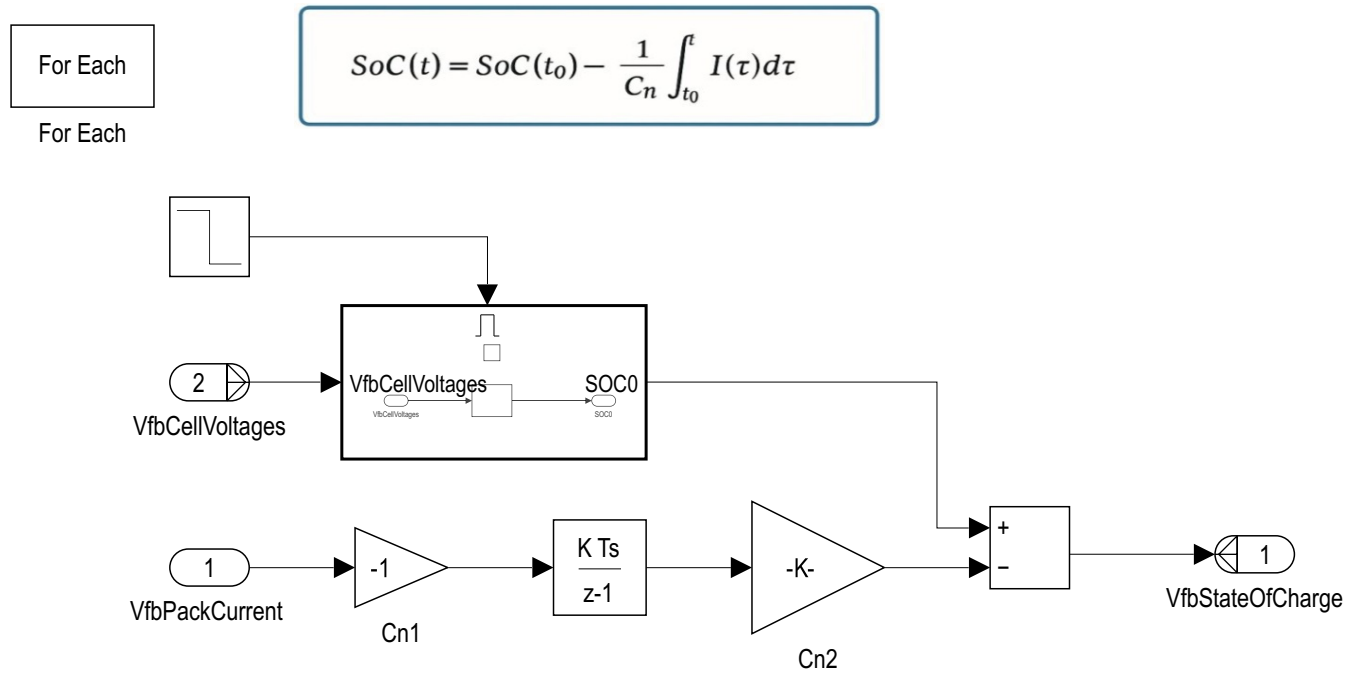

State of Charge Calculation (Coulomb Counting Method)
[CellVoltages]
[PackCurrent]
VfbCellVoltages
VfbPackCurrent
VfbStateOfCharge
[StateOfCharge]
StateOfChargeEstimation

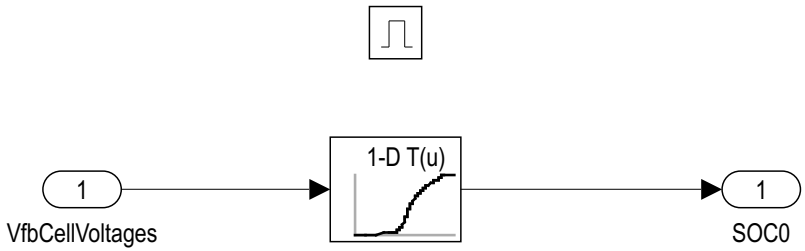

Input Ports
1
VfbStateRequest
2
ChargerOutputVoltage
3
InverterOutputVoltage
4
TotalPackVoltage
5
VfbPackCurrent
6
VfbCellVoltages
7
VfbCellTemperatures
[StateRequest]
[ChargerOutputVoltage]
[InverterOutputVoltage]
[PackVoltage]
[PackCurrent]
[CellVoltages]
[CellTemperatures]


Passive Cell Balancing
[CellVoltages]
CellVoltages
BalanceCommand
BalancingActive
convert
[BalanceCommand]
[BalancingActive]
BalancingLogic


Output Ports
[ChargeCurrentCmd]
[ChargerContactorCommand]
[InverterContactorCommand]
[StateOfCharge]
[BmsOperationalState]
[BalanceCommand]
[BalancingActive]
1
VfbChargeCurrentCmd
2
VfbChargerContactorCommand
3
VfbInverterContactorCommand
4
VfbStateOfCharge
5
VfbBmsOperationalState
6
VfbBalanceCommand
7
VfbBalancingActive


2
VfbPackCurrent
1
VfbCellVoltages
VfbPackCurrent
VfbCellVoltages
VfbStateOfCharge
1
VfbStateOfCharge
StateOfChargeEstimation
(Coulomb Counting)

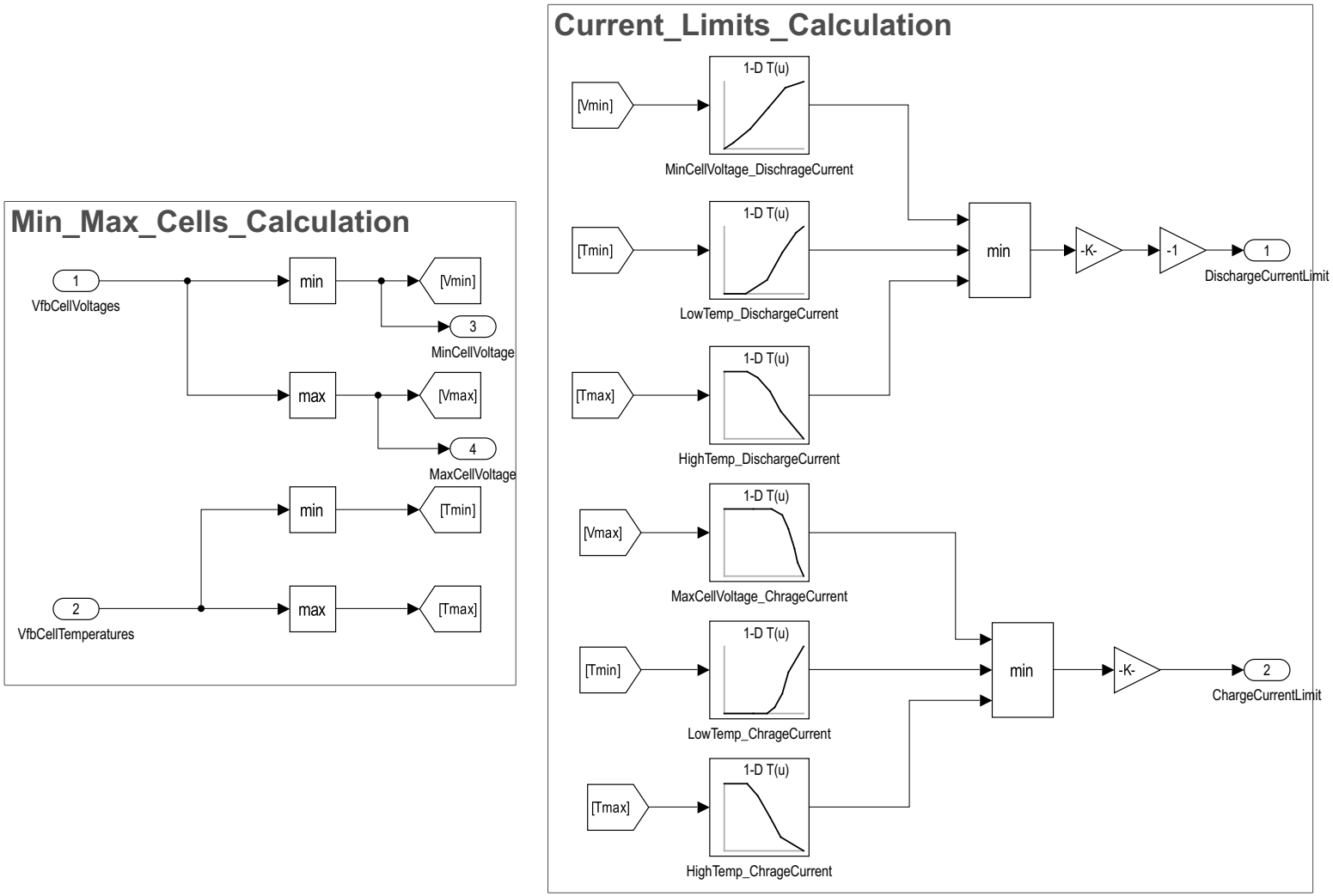

For Each
For Each
$SoC(t) = SoC(t_0) - \frac{1}{C_n}\int_{t_0}^{t} I(\tau)d\tau$
2
VfbCellVoltages
VfbCellVoltages
SOC0
1
VfbPackCurrent
-1
Cn1
K Ts
z-1
-K-
Cn2
+
−
1
VfbStateOfCharge

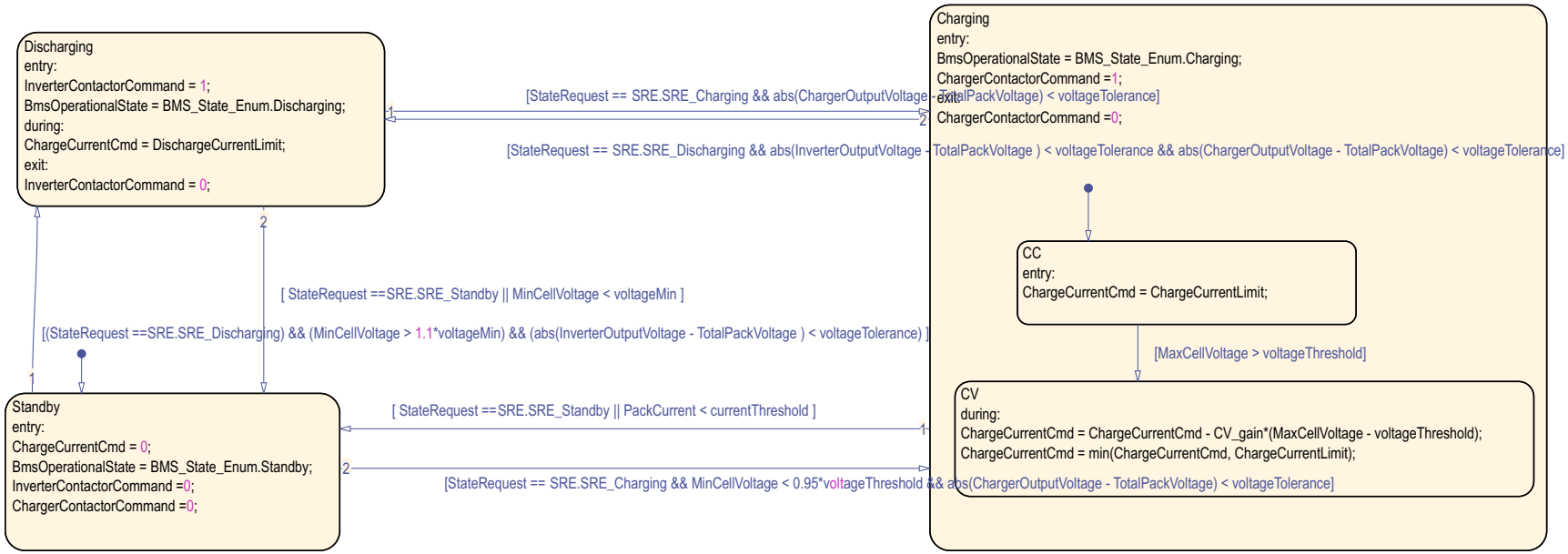

1-D T(u)
1
VfbCellVoltages
1
SOC0

```
function [BalanceCommand,BalancingActive]= fcn(CellVoltages)
Vmin = min(CellVoltages);
BalanceCommand = false(size(CellVoltages));
BalancingActive = false;
deltaV_threshold = 0.05;
for i = 1: length(CellVoltages)
   deltaV = CellVoltages(i)-Vmin;
   BalanceCommand(i) = deltaV >deltaV_threshold;
end

BalancingActive = any(BalanceCommand);

end
```


Min_Max_Cells_Calculation
1
VfbCellVoltages
min
[Vmin]
3
MinCellVoltage
max
[Vmax]
4
MaxCellVoltage
min
[Tmin]
2
VfbCellTemperatures
max
[Tmax]
Current_Limits_Calculation
[Vmin]
1-D T(u)
MinCellVoltage_DischrageCurrent
[Tmin]
1-D T(u)
LowTemp_DischargeCurrent
[Tmax]
1-D T(u)
HighTemp_DischargeCurrent
min
-K-
-1
1
DischargeCurrentLimit
[Vmax]
1-D T(u)
MaxCellVoltage_ChrageCurrent
[Tmin]
1-D T(u)
LowTemp_ChrageCurrent
[Tmax]
1-D T(u)
HighTemp_ChrageCurrent
min
-K-
2
ChargeCurrentLimit


Discharging
entry:
InverterContactorCommand = 1;
BmsOperationalState = BMS_State_Enum.Discharging;
during:
ChargeCurrentCmd = DischargeCurrentLimit;
exit:
InverterContactorCommand = 0;
[StateRequest == SRE.SRE_Charging && abs(ChargerOutputVoltage - TotalPackVoltage) < voltageTolerance]
[StateRequest == SRE.SRE_Discharging && abs(InverterOutputVoltage - TotalPackVoltage) < voltageTolerance && abs(ChargerOutputVoltage - TotalPackVoltage) < voltageTolerance]
Charging
entry:
BmsOperationalState = BMS_State_Enum.Charging;
ChargerContactorCommand =1;
exit:
ChargerContactorCommand =0;
CC
entry:
ChargeCurrentCmd = ChargeCurrentLimit;
[MaxCellVoltage > voltageThreshold]
CV
during:
ChargeCurrentCmd = ChargeCurrentCmd - CV_gain*(MaxCellVoltage - voltageThreshold);
ChargeCurrentCmd = min(ChargeCurrentCmd, ChargeCurrentLimit);
[StateRequest ==SRE.SRE_Standby || MinCellVoltage < voltageMin]
[(StateRequest ==SRE.SRE_Discharging) && (MinCellVoltage > 1.1*voltageMin) && (abs(InverterOutputVoltage - TotalPackVoltage) < voltageTolerance)]
Standby
entry:
ChargeCurrentCmd = 0;
BmsOperationalState = BMS_State_Enum.Standby;
InverterContactorCommand =0;
ChargerContactorCommand =0;
[StateRequest ==SRE.SRE_Standby || PackCurrent < currentThreshold]
[StateRequest == SRE.SRE_Charging && MinCellVoltage < 0.95*voltageThreshold && abs(ChargerOutputVoltage - TotalPackVoltage) < voltageTolerance]